Certificate of Calibration

Standards Laboratory Calibration

Certificate Number WO-01524852

Model Number	33622A
Manufacturer	Keysight Technologies
Description	Waveform generator, 120 MHz, 2-channel
Serial Number	SG53800187
Date of Calibration	30 Sep 2026
Procedure	24BR-TP0900AC3-R2
Temperature	(23.0 ± 1.0) °C
Humidity	(35 to 55) %RH

Customer
Keysight Technologies
Bayan Lepas Free Industrial Zone
Malaysia Sdn Bhd (463532-M)
BAYAN LEPAS Penang 11900
Malaysia

Location of Calibration
Keysight Technologies Inc.
10090 Foothills Blvd.
Roseville CA 95747-7102
UNITED STATES

This certifies that the equipment has been calibrated using applicable Keysight Technologies procedures and accredited to ISO/IEC 17025:2017 and ANSI/NCSL Z540.3-2006. The quality management system is registered to ISO 9001:2015.

As Received Conditions
The measured values of the equipment were observed in specification at the points tested. Additionally, the expanded measurement uncertainty intervals about the measured values were in specification.

Action Taken
- The equipment was adjusted.
- Refer to **Remarks or Special Requirements** Section.

As Completed Conditions
The measured values of the equipment were observed in specification at the points tested. Additionally, the expanded measurement uncertainty intervals about the measured values were in specification.

Remarks or Special Requirements
Additional report as completed only due to STE adjustment.
Refer to additional report for characterized data points.

This calibration report shall not be reproduced, except in full. The documented results relate to the equipment calibrated only.

The test limits stated in the report correspond to the published specifications of the equipment, at the points tested.
This calibration report may refer to equipment manufactured by HP, Agilent and Keysight as being manufactured by Keysight Technologies.
This calibration report is available at Keysight Support Portal http://support.keysight.com/.

Based on the customer's request, the next calibration is due on 30 Mar 2027.

Keysight Technologies Inc
10090 Foothills Blvd.
Roseville CA 95747-7102
UNITED STATES

Dan Allen Roseville Serv. Cntr. Mgr.

Certificate of Calibration

Standards Laboratory Calibration

Certificate Number WO-01524852

Traceability Information

Technician ID 01029259

Measurements are traceable to the International System of Units (SI) via national metrology institutes (www.keysight.com/find/NMI) that are signatories to the CIPM Mutual Recognition Arrangement.

Calibration Equipment Used

Model Number	Model Description	Equipment ID	Cal Due Date
3458A	Digital multimeter, 8.5 digit	SL250275	26 Dec 2026
53131A	Universal Counter, 225 MHz, 10 digit/s, 500 ps. GPIB, RS232	53131A11286	16 Dec 2026
5790A	AC Measurement Standard	SL806123	31 Dec 2026
8491A	Coaxial attenuator, dc-12.4 GHz, Type N	SL257657	31 May 2027
E4419B	Power meter - EPM series, dual channel	E4419B03383	23 Jul 2027
E9304A	Power Sensor-Average, 9 kHz to 6 GHz, -60 to +20 dBm	SL906814	31 May 2027

Traceability Table

	Model	Model Description	Equipment ID	Certificate Number	Trace Value
W,R	3458A	Digital multimeter, 8.5 digit	SL250275	WO-01519364-ANAB:AC-1498	AC Voltage DC Voltage Resistance
W,R	53131A	Universal Counter, 225 MHz, 10 digit/s, 500 ps. GPIB, RS232	53131A11286	WO-01270836-ANAB:AC-1498	Frequency
W,R	5790A	AC Measurement Standard	SL806123	WO-01488392-ANAB:AC-1498	AC Voltage
W,R	8491A	Coaxial attenuator, dc-12.4 GHz, Type N	SL257657	WO-01409475-ANAB:AC-1498	Attenuation
W,R	E4419B	Power meter - EPM series, dual channel	E4419B03383	WO-01469804-ANAB:AC-1498	RF Power
W,R	E9304A	Power Sensor-Average, 9 kHz to 6 GHz, -60 to +20 dBm	SL906814	WO-01409504-ANAB:AC-1498	RF Power

Legend

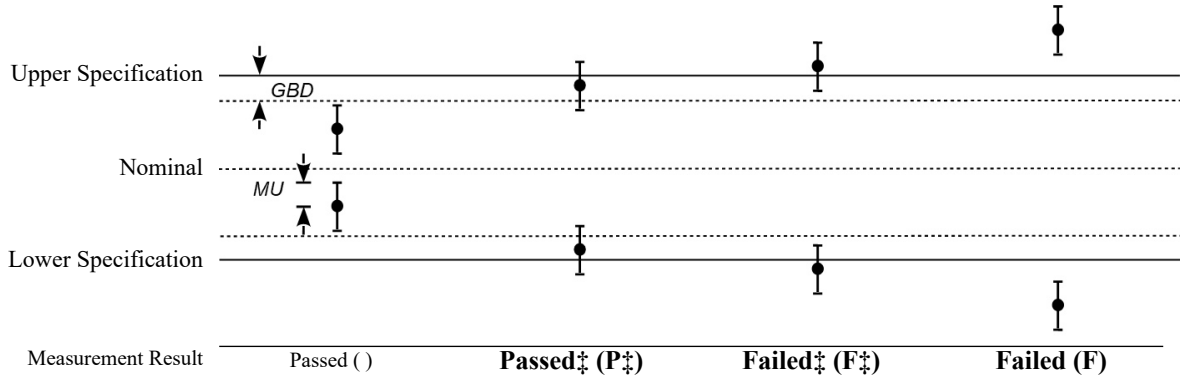
W - Working Standard The calibration equipment used for the calibration of the Model indicated on the first page of the Certificate of calibration.
R - Reference Standard The Reference Standard (Accredited or NMI-calibrated ETE) used to provide traceability to the SI-Units for the calibration parameters listed.

Compliance with Specification

An overall statement of compliance for all tests performed as received, and as completed (if any adjustments / repairs were performed) is included at the beginning of this report. Statements of compliance apply only to warranted specifications. When functional verification tests are performed, results are reported in the "Functional Test" section, and do not affect these statements of compliance.
The status summaries relate to the tested item only. A final decision about whether the item's performance actually satisfies requirements of the user can only be made by the user.

Measurement results are reported as:

- Passed ( ) - The measured values of the equipment were observed in specification at the points tested. Additionally, the expanded measurement uncertainty intervals about the measured values were in specification.
- Passed‡ (P‡) - The measured values of the equipment were observed in specification at the points tested. However, a portion of the expanded measurement uncertainty intervals about one or more measured values exceeded specification. Consequently, compliance with specification cannot be declared based on the stated coverage probability.
- Failed‡ (F‡) - One or more measured values of the equipment were observed out of specification at the points tested. However, a portion of the expanded measurement uncertainty intervals about one or more measured values were in specification. Consequently, non-compliance with specification cannot be declared based on the stated coverage probability.
- Failed (F) - One or more measured values of the equipment were observed out of specification at the points tested. Additionally, the expanded measurement uncertainty intervals about one or more measured values were entirely outside the specification.



MU = 95% expanded measurement uncertainty.
( ) This result is indicated on the measurement report as a blank space in the column labeled "Status" or "Sts".
Note: For more information on the level of risk such as false accept and false reject and statistical assumptions of these statements of conformity, please visit: www.keysight.com/find/decisionrules.

Uncertainty of Measurement

The uncertainty evaluation has been performed in accordance with ISO/IEC Guide 98-3:2008 (GUM). The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k such that the coverage probability corresponds to approximately 95%. This probability corresponds to a coverage factor of k=2 for a normal distribution.

Certificate of Calibration

Standards Laboratory Calibration

Certificate Number WO-01524852

Calibration Test Results Summary

Calibration test results of warranted specifications

Test Name	As Received Status	As Completed Status
Internal Timebase	Passed	Passed
AC Amplitude	Passed	Passed
DC Offset Voltage	Passed	Passed
1 Vpp Range Flatness	Passed	Passed
4 Vpp Range Flatness	Passed	Passed
8 Vpp Range Flatness	Passed	Passed

Functional Test Results Summary

The following functional test results are not part of an accredited delivery, even if they are part of an otherwise accredited calibration report.

The following tests document the functional verification of the instruments' non-warranted performance. Neither a statement of conformance or decision rule is used for a Functional Test, measurement uncertainties are only provided by exception. For a "Functional Test" the test results are reported as "As Expected" when showing expected performance and "Not As Expected" otherwise. "As Expected" results of individual test points are indicated in the measurement report by a blank space in the column labeled "Status" to allow easier recognition of any "Not As Expected" points. If a functional test result is reported as "Not As Expected", repair and/or adjustment is recommended. Test results reported as "Done" are possible if no limits are applied. For qualitative or quantitative "Functional Tests" the test results are not warranted, and no judgment is made. The "actual" measured results are helpful to users for some applications.

Test Name	As Received Status	As Completed Status
Self-Test	As Expected	As Expected

Tested Configuration

Firmware Version	A.02.03-2.40-03-64-02
(As Rec)	A.02.03-2.40-03-64-02

Internal Timebase — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
10 MHz SINE WAVE	−10.00 Hz	−3.01 Hz	10.00 Hz	1.9 Hz	

AC Amplitude — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Frequency 1 kHz					
Channel1 AutoRanging off					
0.4 V	0.395293 V	0.397469 V	0.404707 V	0.00018 V	
Channel1 AutoRanging on					
0.4 V	0.395293 V	0.397448 V	0.404707 V	0.00010 V	
1.0 mV	0.000283 V	0.000996 V	0.001717 V	0.0000035 V	
4.0 mV	0.003253 V	0.003975 V	0.004747 V	0.0000039 V	
10.0 mV	0.009193 V	0.009938 V	0.010807 V	0.0000038 V	
20.0 mV	0.019093 V	0.019881 V	0.020907 V	0.0000042 V	
40.0 mV	0.038893 V	0.039773 V	0.041107 V	0.0000060 V	
100 mV	0.098293 V	0.099422 V	0.101707 V	0.000012 V	
1.0 V	0.989293 V	0.993680 V	1.010707 V	0.00012 V	
2.5 V	2.474293 V	2.484047 V	2.525707 V	0.00046 V	
7.0 V	6.929293 V	6.955791 V	7.070707 V	0.00086 V	
Channel2 AutoRanging off					
0.4 V	0.395293 V	0.397475 V	0.404707 V	0.00018 V	
Channel2 AutoRanging on					
0.4 V	0.395293 V	0.397501 V	0.404707 V	0.00010 V	
1.0 mV	0.000283 V	0.000993 V	0.001717 V	0.0000035 V	
4.0 mV	0.003253 V	0.003972 V	0.004747 V	0.0000039 V	
10.0 mV	0.009193 V	0.009933 V	0.010807 V	0.0000038 V	
20.0 mV	0.019093 V	0.019878 V	0.020907 V	0.0000042 V	
40.0 mV	0.038893 V	0.039762 V	0.041107 V	0.0000060 V	
100 mV	0.098293 V	0.099403 V	0.101707 V	0.000012 V	
1.0 V	0.989293 V	0.993788 V	1.010707 V	0.00012 V	
2.5 V	2.474293 V	2.484408 V	2.525707 V	0.00046 V	
7.0 V	6.929293 V	6.956225 V	7.070707 V	0.00086 V	

DC Offset Voltage — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
0.0 V	−0.0020 V	0.0001 V	0.0020 V	5.9E-005 V	
0.5 V	0.4930 V	0.4999 V	0.5070 V	0.00014 V	
10.0 V	9.8980 V	9.9950 V	10.1020 V	0.00029 V	
Channel 2					
0.0 V	−0.0020 V	0.0001 V	0.0020 V	5.9E-005 V	
0.5 V	0.4930 V	0.5002 V	0.5070 V	0.00014 V	
10.0 V	9.8980 V	10.0033 V	10.1020 V	0.00029 V	

1 Vpp Range Flatness — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
1 kHz		Ref			
100 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.011 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.017 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.017 dB	0.100 dB	0.026 dB	
10 MHz	−0.200 dB	0.029 dB	0.200 dB	0.029 dB	
20 MHz	−0.200 dB	0.033 dB	0.200 dB	0.031 dB	
30 MHz	−0.200 dB	0.030 dB	0.200 dB	0.030 dB	
40 MHz	−0.200 dB	0.047 dB	0.200 dB	0.030 dB	
50 MHz	−0.200 dB	0.060 dB	0.200 dB	0.021 dB	
60 MHz	−0.200 dB	0.061 dB	0.200 dB	0.027 dB	
70 MHz	−0.300 dB	0.066 dB	0.300 dB	0.027 dB	
80 MHz	−0.300 dB	0.085 dB	0.300 dB	0.031 dB	
90 MHz	−0.400 dB	0.111 dB	0.400 dB	0.031 dB	
100 MHz	−0.400 dB	0.098 dB	0.400 dB	0.031 dB	
110 MHz	−0.400 dB	0.097 dB	0.400 dB	0.031 dB	
120 MHz	−0.400 dB	0.102 dB	0.400 dB	0.031 dB	
Channel 2					
1 kHz		Ref			
100 kHz	−0.100 dB	0.008 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.009 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.011 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.011 dB	0.100 dB	0.026 dB	
10 MHz	−0.200 dB	0.023 dB	0.200 dB	0.029 dB	
20 MHz	−0.200 dB	0.032 dB	0.200 dB	0.031 dB	
30 MHz	−0.200 dB	0.030 dB	0.200 dB	0.030 dB	
40 MHz	−0.200 dB	0.040 dB	0.200 dB	0.030 dB	
50 MHz	−0.200 dB	0.044 dB	0.200 dB	0.021 dB	

1 Vpp Range Flatness (cont.)

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
60 MHz	−0.200 dB	0.052 dB	0.200 dB	0.027 dB	
70 MHz	−0.300 dB	0.061 dB	0.300 dB	0.027 dB	
80 MHz	−0.300 dB	0.086 dB	0.300 dB	0.031 dB	
90 MHz	−0.400 dB	0.122 dB	0.400 dB	0.031 dB	
100 MHz	−0.400 dB	0.126 dB	0.400 dB	0.031 dB	
110 MHz	−0.400 dB	0.122 dB	0.400 dB	0.031 dB	
120 MHz	−0.400 dB	0.135 dB	0.400 dB	0.031 dB	

4 Vpp Range Flatness **Passed**

TEST COND.	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
1 kHz		Ref			
100 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.004 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	0.013 dB	0.250 dB	0.029 dB	
20 MHz	−0.250 dB	0.026 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	0.031 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	0.043 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	0.049 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	0.035 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	0.027 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	0.032 dB	0.400 dB	0.031 dB	
90 MHz	−0.500 dB	0.032 dB	0.500 dB	0.031 dB	
100 MHz	−0.500 dB	0.000 dB	0.500 dB	0.031 dB	
110 MHz	−0.500 dB	−0.013 dB	0.500 dB	0.031 dB	
120 MHz	−0.500 dB	−0.014 dB	0.500 dB	0.031 dB	
Channel 2					
1 kHz		Ref			
100 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	0.017 dB	0.250 dB	0.029 dB	
20 MHz	−0.250 dB	0.032 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	0.035 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	0.041 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	0.038 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	0.029 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	0.024 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	0.032 dB	0.400 dB	0.031 dB	
90 MHz	−0.500 dB	0.043 dB	0.500 dB	0.031 dB	
100 MHz	−0.500 dB	0.025 dB	0.500 dB	0.031 dB	
110 MHz	−0.500 dB	0.007 dB	0.500 dB	0.032 dB	

4 Vpp Range Flatness (cont.)

TEST COND.	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
120 MHz	−0.500 dB	0.017 dB	0.500 dB	0.032 dB	

8 Vpp Range Flatness — Passed

TEST COND.	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
1 kHz		Ref			
100 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.014 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.018 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.006 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	0.011 dB	0.250 dB	0.030 dB	
20 MHz	−0.250 dB	0.017 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	0.001 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	0.022 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	0.031 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	0.030 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	0.010 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	0.084 dB	0.400 dB	0.031 dB	
Channel 2					
1 kHz		Ref			
100 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.013 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.016 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.018 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	0.028 dB	0.250 dB	0.030 dB	
20 MHz	−0.250 dB	0.027 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	0.018 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	0.028 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	0.021 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	0.028 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	0.015 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	0.095 dB	0.400 dB	0.030 dB	

Self-Test — As Expected

TEST COND.	RESULT	Status
SELF TEST	DONE	

Internal Timebase — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
10 MHz SINE WAVE	−10.00 Hz	0.01 Hz	10.00 Hz	1.9 Hz	

AC Amplitude — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Frequency 1 kHz					
Channel1 AutoRanging off					
0.4 V	0.395293 V	0.397764 V	0.404707 V	0.00018 V	
Channel1 AutoRanging on					
0.4 V	0.395293 V	0.397706 V	0.404707 V	0.00010 V	
1.0 mV	0.000283 V	0.000995 V	0.001717 V	0.0000035 V	
4.0 mV	0.003253 V	0.003977 V	0.004747 V	0.0000039 V	
10.0 mV	0.009193 V	0.009942 V	0.010807 V	0.0000038 V	
20.0 mV	0.019093 V	0.019887 V	0.020907 V	0.0000042 V	
40.0 mV	0.038893 V	0.039778 V	0.041107 V	0.0000060 V	
100 mV	0.098293 V	0.099440 V	0.101707 V	0.000012 V	
1.0 V	0.989293 V	0.994153 V	1.010707 V	0.00012 V	
2.5 V	2.474293 V	2.485288 V	2.525707 V	0.00046 V	
7.0 V	6.929293 V	6.959990 V	7.070707 V	0.00086 V	
Channel2 AutoRanging off					
0.4 V	0.395293 V	0.397689 V	0.404707 V	0.00018 V	
Channel2 AutoRanging on					
0.4 V	0.395293 V	0.397719 V	0.404707 V	0.00010 V	
1.0 mV	0.000283 V	0.000994 V	0.001717 V	0.0000035 V	
4.0 mV	0.003253 V	0.003975 V	0.004747 V	0.0000039 V	
10.0 mV	0.009193 V	0.009944 V	0.010807 V	0.0000038 V	
20.0 mV	0.019093 V	0.019890 V	0.020907 V	0.0000042 V	
40.0 mV	0.038893 V	0.039777 V	0.041107 V	0.0000060 V	
100 mV	0.098293 V	0.099441 V	0.101707 V	0.000012 V	
1.0 V	0.989293 V	0.994400 V	1.010707 V	0.00012 V	
2.5 V	2.474293 V	2.485943 V	2.525707 V	0.00046 V	
7.0 V	6.929293 V	6.960611 V	7.070707 V	0.00086 V	

DC Offset Voltage — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
0.0 V	−0.0020 V	0.0000 V	0.0020 V	5.9E-005 V	
0.5 V	0.4930 V	0.5003 V	0.5070 V	0.00014 V	
10.0 V	9.8980 V	10.0017 V	10.1020 V	0.00029 V	
Channel 2					
0.0 V	−0.0020 V	0.0000 V	0.0020 V	5.9E-005 V	
0.5 V	0.4930 V	0.5003 V	0.5070 V	0.00014 V	
10.0 V	9.8980 V	10.0035 V	10.1020 V	0.00029 V	

1 Vpp Range Flatness — Passed

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
1 kHz		Ref			
100 kHz	−0.100 dB	−0.010 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	−0.003 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	−0.004 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	−0.006 dB	0.100 dB	0.026 dB	
10 MHz	−0.200 dB	−0.004 dB	0.200 dB	0.029 dB	
20 MHz	−0.200 dB	−0.004 dB	0.200 dB	0.031 dB	
30 MHz	−0.200 dB	−0.006 dB	0.200 dB	0.030 dB	
40 MHz	−0.200 dB	−0.010 dB	0.200 dB	0.030 dB	
50 MHz	−0.200 dB	−0.010 dB	0.200 dB	0.021 dB	
60 MHz	−0.200 dB	−0.007 dB	0.200 dB	0.027 dB	
70 MHz	−0.300 dB	0.000 dB	0.300 dB	0.027 dB	
80 MHz	−0.300 dB	0.009 dB	0.300 dB	0.031 dB	
90 MHz	−0.400 dB	0.019 dB	0.400 dB	0.031 dB	
100 MHz	−0.400 dB	0.028 dB	0.400 dB	0.031 dB	
110 MHz	−0.400 dB	0.031 dB	0.400 dB	0.031 dB	
120 MHz	−0.400 dB	0.020 dB	0.400 dB	0.031 dB	
Channel 2					
1 kHz		Ref			
100 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	−0.005 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	−0.005 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	−0.007 dB	0.100 dB	0.026 dB	
10 MHz	−0.200 dB	−0.005 dB	0.200 dB	0.029 dB	
20 MHz	−0.200 dB	−0.003 dB	0.200 dB	0.031 dB	
30 MHz	−0.200 dB	−0.013 dB	0.200 dB	0.030 dB	
40 MHz	−0.200 dB	−0.012 dB	0.200 dB	0.030 dB	
50 MHz	−0.200 dB	−0.010 dB	0.200 dB	0.021 dB	

1 Vpp Range Flatness (cont.)

TEST CONDITIONS	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
60 MHz	−0.200 dB	−0.010 dB	0.200 dB	0.027 dB	
70 MHz	−0.300 dB	−0.002 dB	0.300 dB	0.027 dB	
80 MHz	−0.300 dB	0.008 dB	0.300 dB	0.031 dB	
90 MHz	−0.400 dB	0.022 dB	0.400 dB	0.031 dB	
100 MHz	−0.400 dB	0.036 dB	0.400 dB	0.031 dB	
110 MHz	−0.400 dB	0.045 dB	0.400 dB	0.031 dB	
120 MHz	−0.400 dB	0.043 dB	0.400 dB	0.031 dB	

4 Vpp Range Flatness

Passed

TEST COND.	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
1 kHz		Ref			
100 kHz	−0.100 dB	0.007 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.015 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.015 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.014 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	0.010 dB	0.250 dB	0.029 dB	
20 MHz	−0.250 dB	0.014 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	0.014 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	0.005 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	−0.006 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	−0.017 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	−0.031 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	−0.044 dB	0.400 dB	0.031 dB	
90 MHz	−0.500 dB	−0.055 dB	0.500 dB	0.031 dB	
100 MHz	−0.500 dB	−0.063 dB	0.500 dB	0.031 dB	
110 MHz	−0.500 dB	−0.072 dB	0.500 dB	0.031 dB	
120 MHz	−0.500 dB	−0.088 dB	0.500 dB	0.031 dB	
Channel 2					
1 kHz		Ref			
100 kHz	−0.100 dB	0.008 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	−0.014 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	−0.037 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.011 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	0.020 dB	0.250 dB	0.029 dB	
20 MHz	−0.250 dB	0.028 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	0.011 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	−0.012 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	−0.014 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	−0.027 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	−0.038 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	−0.053 dB	0.400 dB	0.031 dB	
90 MHz	−0.500 dB	−0.057 dB	0.500 dB	0.031 dB	
100 MHz	−0.500 dB	−0.060 dB	0.500 dB	0.031 dB	
110 MHz	−0.500 dB	−0.064 dB	0.500 dB	0.032 dB	

4 Vpp Range Flatness (cont.)

TEST COND.	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
120 MHz	−0.500 dB	−0.074 dB	0.500 dB	0.032 dB	

8 Vpp Range Flatness — Passed

TEST COND.	MINIMUM	MEASURED	MAXIMUM	UNCERT.	Status
Channel 1					
1 kHz		Ref			
100 kHz	−0.100 dB	0.011 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	0.004 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	0.016 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	0.008 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	0.010 dB	0.250 dB	0.030 dB	
20 MHz	−0.250 dB	0.011 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	0.008 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	0.001 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	0.001 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	0.005 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	0.021 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	0.072 dB	0.400 dB	0.031 dB	
Channel 2					
1 kHz		Ref			
100 kHz	−0.100 dB	0.008 dB	0.100 dB	0.026 dB	
500 kHz	−0.100 dB	−0.002 dB	0.100 dB	0.026 dB	
1 MHz	−0.100 dB	−0.002 dB	0.100 dB	0.026 dB	
5 MHz	−0.100 dB	−0.003 dB	0.100 dB	0.026 dB	
10 MHz	−0.250 dB	−0.004 dB	0.250 dB	0.030 dB	
20 MHz	−0.250 dB	−0.004 dB	0.250 dB	0.031 dB	
30 MHz	−0.250 dB	−0.005 dB	0.250 dB	0.030 dB	
40 MHz	−0.250 dB	−0.010 dB	0.250 dB	0.030 dB	
50 MHz	−0.250 dB	−0.011 dB	0.250 dB	0.021 dB	
60 MHz	−0.250 dB	−0.007 dB	0.250 dB	0.027 dB	
70 MHz	−0.400 dB	0.008 dB	0.400 dB	0.027 dB	
80 MHz	−0.400 dB	0.056 dB	0.400 dB	0.030 dB	

Self-Test — As Expected

TEST COND.	RESULT	Status
SELF TEST	DONE	

KEYSIGHT TECHNOLOGIES

Keysight Technologies Primary Standards Lab · 10090 Foothills Blvd · Roseville, CA 95747

Standards Laboratory
Calibration Report

Model #	33622A	**Certificate #**	WO-01524852
Serial Number	SG53800187		
Asset #	SL906151	**Cal Date**	30 Sep 2026

As Completed Measurement Data

Flatness Characterization referenced to 100kHz of nominal voltage listed for automated 3458A High Frequency and SCAL programs. Channel 1 characterized only. All measurements are characterized using the Wideband connection of a Fluke 5790A with a matched & identified cable assigned to this 33622A.
Values listed are valid at the end of the assigned cable with 50 Ω impedance matching.

Test Points (RMS)	33622 Flatness Reading	5790A Correction	DUT Flatness	cf Data
10mV @ 4MHz	-0.299	0.0439	-0.2864	-0.02491
100mV @ 4MHz	-0.130	0.0382	-0.0873	-0.00758
100mV @ 8MHz	-0.290	0.0576	-0.2279	-0.01981
100mV @ 10MHz	-0.304	0.0667	-0.2328	-0.02024
1V @ 4MHz	-0.155	-0.0498	-0.1163	-0.01010
1V @ 8MHz	-0.289	-0.0535	-0.2545	-0.02213
1V @ 10MHz	-0.301	-0.0635	-0.2763	-0.02403
3V @ 2MHz	-0.316	-0.0605	-0.3121	-0.02715
3V @ 4MHz	-0.360	-0.0318	-0.3272	-0.02846
3V @ 8MHz	-0.486	-0.0202	-0.4415	-0.03843
3V @ 10MHz	-0.498	-0.0214	-0.4544	-0.03955

Characterized Frequency	Average Error for Freq. in %	Error corrected to dB value	0.4% Flatness Check
2MHz	-0.3121 %	-0.027 dB	PASS
4MHz	-0.2043 %	-0.018 dB	PASS
8MHz	-0.2456 %	-0.021 dB	PASS
10MHz	-0.3212 %	-0.028 dB	PASS

Acceptance criteria is based on flatness for each frequency tested and is not related to the average error for each frequency.

This report shall not be reproduced except in full, without written approval of the Keysight Technologies Primary Standards Lab.